

EXPERT CONSENSUS SYSTEMS OF CARE DOCUMENT

Operator and Institutional Recommendations and Requirements for Tricuspid Interventions: 2026 ACC/AHA/ASE/HRS/STS Expert Consensus Systems of Care Document

A Joint Report of the American College of Cardiology, the American Heart Association, the American Society of Echocardiography, the Heart Rhythm Society, and the Society of Thoracic Surgeons

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TABLE OF CONTENTS

PREAMBLE	■	1.2. Process	■
1. INTRODUCTION	■	2. METHODS	■
1.1. Background	■	2.1. Abbreviations	■

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3. PATIENT EVALUATION AND SELECTION	■
3.1. Multidisciplinary Team	■
3.1.1. Knowledge Base and Skills of the MDT	■
3.1.2. Structure and Function of the MDT	■
3.2. SDM Requirements	■
3.3. Facilities and Institutional Requirements	■
3.4. Outcome Requirements	■
3.5. National Registry Requirements	■
3.6. Requirements for Establishing and Maintaining a TTVI Program	■
3.6.1. Requirements for the Intraoperative Technical Aspects of T-TEER and TTVR	■
3.6.2. Qualifications to Maintain a T-TEER or TTVR Program	■
3.7. Compliance With Document Recommendations	■
CONCLUSIONS	■
REFERENCES	■
APPENDIX 1	
Author Relationships With Industry and Other Entities (Relevant)	■
APPENDIX 2	
Peer Reviewer Relationships With Industry and Other Entities (Comprehensive)	■

PREAMBLE

Transcatheter valve repair and replacement procedures have evolved considerably over the past 2 decades. Transcatheter aortic valve implantation has become a reasonable alternative to surgical aortic valve replacement for the management of appropriately selected adult patients with severe aortic stenosis across the surgical risk spectrum. Mitral transcatheter edge-to-edge repair (M-TEER) has demonstrated use in the management of symptomatic (New York Heart Association [NYHA] class III-IV) patients with severe degenerative mitral regurgitation who are deemed to be poor candidates for surgery and patients with severe secondary mitral regurgitation with reduced left ventricular ejection fraction and persistent heart failure (HF) symptoms despite optimal medical and device therapy.¹ In 2024, the U.S. Food and Drug Administration (FDA) approved both a transcatheter tricuspid valve replacement (TTVR) system (EVOQUE, Edwards Lifesciences) and a transcatheter tricuspid valve edge-to-edge repair (T-TEER) system (TriClip, Abbott) for use in the management of patients with symptomatic

severe tricuspid regurgitation (TR) despite optimal medical therapy (OMT) and for whom tricuspid valve (TV) intervention is deemed appropriate by a multidisciplinary heart team (MDT).^{2,3} Approval of these devices was targeted specifically to improve patient health status as reflected in quality of life (QoL) metrics and followed publication of the results of pivotal randomized controlled trials (RCTs) that assessed the effectiveness and safety of transcatheter tricuspid valve intervention (TTVI) plus OMT versus OMT alone.^{4,5}

To date, the use of these devices has not been shown to impact rates of survival. Evidence regarding other outcomes (eg, hospitalization for HF, TV surgery) beyond the time points designated for adjudication of the primary endpoint in the study protocols for the pivotal RCTs is evolving (see Section 1.1).⁶ The Centers for Medicare & Medicaid Services (CMS) announced that it will cover TTVR and T-TEER for the treatment of symptomatic TR under the Coverage with Evidence Development (CED).^{7,8} The CED study criteria include the primary outcomes of all-cause death, hospitalizations, or a composite of these, through a minimum of 24 months. In addition, for composite outcome measures, individual physiologic, patient-reported, and other relevant health outcomes should be directionally consistent. Outcomes are to be analyzed against an active comparator. This CED pathway differs from those applied to transcatheter aortic valve implantation and M-TEER; it does not specify institutional or operator case volumes but stipulates that patients are to be under the care of an MDT.

Experience gained during the conduct of RCTs,^{4,5,9} single-arm studies,^{10,11} and real-world registries^{12,13} has underscored the critical role of a cohesive and highly functional MDT, inclusive of an expert interventional echocardiographer and, when necessary, a cardiac implantable electronic device (CIED) specialist. The MDT learning curve must account for appropriate patient selection, findings on multimodality imaging assessment, appreciation of contemporary outcomes with surgical treatment of TR, optimal HF and rhythm management strategies, iterative improvements in device and imaging technology, operator and interventional echocardiographic skills, organizational experience, and patient-centered shared decision-making (SDM).¹⁴ The TTVI space is evolving rapidly; it is anticipated that additional transcatheter systems for treatment of severe TR will likely gain FDA approval in the future, necessitating periodic updates to this document. A second T-TEER system (PASCAL, Edwards Lifesciences) received European CE Mark approval in 2020.

The American College of Cardiology (ACC), American Heart Association (AHA), American Society of Echocardiography (ASE), Heart Rhythm Society (HRS), and Society of Thoracic Surgeons (STS) joined to provide guidance

and criteria for operators and institutions to evaluate and benchmark their ability to initiate, sustain, or both a TV intervention program. The ACC, AHA, ASE, HRS, and STS believe that adhering to these recommendations will increase the likelihood of the safe and effective incorporation of these therapies into the management of patients with severe TR. The recommendations seek to balance achieving the best quality outcomes with ensuring access to cutting-edge therapies after FDA approval. Quality and safety concerns reflect the complex and multidisciplinary nature of planning and performing TTVIs, recognized hazards, and the relatively limited randomized trial evidence base supporting their efficacy. A recently released multidisciplinary ACC companion document, “10 Issues for the Clinician in Tricuspid Regurgitation Evaluation and Management: 2025 ACC Expert Consensus Decision Pathway,”¹⁵ can be consulted for additional background on the evaluation and management of patients with severe TR pending anticipated future updates to the “2020 ACC/AHA Guideline for the Management of Patients With Valvular Heart Disease.”¹

This document focuses on the recommended components of a TTVI program for the treatment of native-valve severe TR using FDA-approved devices for this indication. It does not address the use of transcatheter aortic valves for treatment of either bioprosthetic TV failure (valve-in-valve) or recurrent severe TR after surgical TV annuloplasty (valve-in-ring). However, it is reasonable to assume that a well-functioning MDT performing TTVI would also consider these therapies when appropriate.

1. INTRODUCTION

1.1. Background

The causes of TR fall broadly into 1 of 3 categories (Table 1).¹⁶ Primary TR refers to disease affecting the TV leaflets or chordae, examples of which include infective endocarditis, prolapse, rheumatic heart disease, radiation, carcinoid, and Ebstein's anomaly. Secondary TR develops as a consequence of right-sided chamber remodeling, tricuspid annular dilation, and leaflet tethering, most often in the context of left-sided valve or myocardial disease, right ventricular (RV) dysfunction, pulmonary hypertension, atrial fibrillation (AF), or a combination of these. Secondary TR due to RV remodeling is referred to as *ventricular-secondary TR*, whereas the term *atrial secondary TR* applies to patients with persistent AF, HF with preserved ejection fraction, or both in whom right atrial remodeling and tricuspid annular enlargement dominate the clinical phenotype without significant RV dilation.^{17,18} Mixed secondary TR with elements of both ventricular and atrial remodeling is frequently encountered.¹⁹ A third category for TR applies to patients with CIED-related TR, which may result from

TABLE 1 Etiologic Classification of TR

Classification	Subclassification	Etiologies
Primary TR		■ Degenerative (prolapse, flail) ■ Infective endocarditis ■ Trauma to the chest wall or leaflets ■ Ebstein's anomaly ■ Carcinoid disease ■ Mediastinal radiation ■ Rheumatic disease
Secondary TR	V-STR	PH attributable to: ■ LV systolic dysfunction or left-sided valve disease ■ Pulmonary arterial hypertension ■ Chronic lung disease ■ Chronic pulmonary thromboembolic disease ■ Intracardiac shunt
	A-STR	■ Atrial arrhythmias ■ HFpEF
CIED-related TR	CIED-related causative TR	■ Direct interaction of the lead with the TV leaflets or subvalvular apparatus
	CIED-related incidental TR	■ CIED present but not the direct cause of TR

Reprinted with permission from Hahn et al.³⁶

A-STR = atrial secondary tricuspid regurgitation; CIED = cardiac implantable electronic device; HFpEF = heart failure with preserved ejection fraction; LV = left ventricular; PH = pulmonary hypertension; TR = tricuspid regurgitation; TV = transcatheter valve; V-STR = ventricular secondary tricuspid regurgitation.

interaction of the CIED lead with any portion of the TV apparatus (eg, leaflet, chordae, or papillary muscle); valve fibrosis related to contact with the lead; dyssynchronous RV pacing; or leaflet avulsion at time of lead removal. Most TR cases encountered clinically are secondary or CIED-related.²⁰

Increasing grades of TR severity are associated with worsening HF symptoms, progressive signs of low cardiac output and systemic congestion, and reduced event-free survival.^{21,22} The typical symptoms and physical examination findings associated with significant TR are summarized in the companion 2025 Expert Consensus Decision Pathway.¹⁵ Transthoracic echocardiography is the imaging modality of first choice to define the cause and severity of TR, as recommended by the ASE and the Tricuspid Valve Academic Research Consortium^{17,23,24} using a combination of structural, qualitative, semiquantitative, and quantitative measures. Advanced imaging with transesophageal echocardiography (TEE), cardiac magnetic resonance, and cardiac computed tomography is used selectively for more refined assessment of TV and RV anatomy and function. TEE is required to complete planning for TV intervention and for intraprocedural guidance. Intracardiac echocardiography may be used for intraprocedural imaging, and cardiac computed tomography is a component of planning for TTVR.

Guideline-directed medical therapy and MDT-supervised HF, rhythm, and CIED management are foundational to the care of patients with severe TR. TV

intervention is reserved for patients with severe TR who remain symptomatic despite these combined efforts. Whereas TV surgery is recommended at the time of left-sided valve surgery for patients with severe TR, the timing of referral for surgical treatment of isolated TR is less well defined. With the exception of endocarditis, surgery has historically been considered late in the natural history of the disease because of delayed recognition and patient comorbidities. Isolated TV surgery can be beneficial to reduce symptoms and recurrent hospitalizations in patients with signs and symptoms of right-sided HF and severe secondary TR who are poorly responsive to medical therapy. TV surgery can also be considered for selected patients who have undergone prior cardiac surgery and who have progressive right-sided HF poorly responsive to medical therapy.¹ The increased risks of surgery in the latter context should be weighed carefully by the MDT. Outcomes after isolated TV surgery have improved with better patient selection and preoperative optimization of volume status, along with hepatic and renal function. Using data from the National (Nationwide) Inpatient Sample, Zack et al²⁵ reported an in-hospital death rate of 8.8% from 2004 to 2013 for isolated TV replacement or repair (n = 5005; median age, 62 years). In an analysis using the STS Adult Cardiac Surgery Database of 13,587 patients undergoing isolated TV surgery from 2017 to 2023, Thourani et al²⁶ reported an early death rate of 5.6%. Over the course of study, the rates of operative death, stroke, renal failure, reoperation, combined morbidity and death, prolonged ventilation, and length of stay decreased. This analysis led to the development of a new risk model for 8 outcomes after isolated TV surgery, which can aid in patient counseling, clinical decision-making, quality improvement, and programmatic benchmarking. Other surgical risk scores can be consulted.^{27,28} In view of past reports, the long-term durability of surgical tricuspid valve repair using current techniques warrants investigation.²⁹ Randomized trials of surgical therapy versus either medical therapy or transcatheter intervention for severe TR have not been done.

T-TEER is based on the surgical Alfieri repair procedure and mimics what has been accomplished for M-TEER. The TRILUMINATE Pivotal (Clinical Trial to Evaluate Cardiovascular Outcomes in Patients Treated With the Tricuspid Valve Repair System Pivotal) trial was an international, open label RCT of T-TEER using the TriClip/TriClip G4 system plus OMT (n = 175) versus OMT alone (n = 175).⁴ Mean age was 78 years, 54.9% were women, 6.6% had undergone prior surgical mitral valve repair, and 11.4% had undergone prior M-TEER. The primary endpoint, a hierarchical composite of all-cause death or TV surgery, heart failure hospitalization (HFH), and improvement in QoL scores measured by the Kansas City Cardiomyopathy

Questionnaire score at 1 year, favored T-TEER (win ratio: 1.48; 95% CI: 1.06–2.13; $P = 0.02$) driven primarily by an improvement in QoL scores. At 1 year, the Kansas City Cardiomyopathy Questionnaire score increased by a mean of 12.3 points in the T-TEER group versus 0.6 points in the control group ($P < 0.001$). The rates of all-cause death or TV surgery (9.4% versus 10.6%) and HFH (0.21 versus 0.17 events/patient year) did not differ between the 2 arms. The proportion of patients experiencing TR reduction to moderate or less severity at 30 days was superior with T-TEER (87% versus 4.8%; $P < 0.001$). The mean increase in the Kansas City Cardiomyopathy Questionnaire score at 1-year follow-up was greater among those who achieved a greater degree of durable TR reduction (15.6 among patients with moderate or less residual TR at 1 year versus 3.8 among patients with severe or greater residual TR). Change in 6-minute walk distance from baseline to 1 year did not differ between treatment groups nor did the intensity of diuretic therapy. No excess hazard was observed of the need for permanent pacemaker implantation. The rate of severe bleeding was 5.2% within 1 year in the T-TEER arm. At 2 years, the degree of TR reduction with T-TEER appeared durable (84% randomized to T-TEER had moderate or less TR). In addition, the annualized rate of recurrent HFH was significantly lower in the T-TEER group compared with controls (0.19 versus 0.26 events/patient-year; $P = 0.02$); the cumulative function curves began to diverge after 1 year. No significant difference was observed in the 2-year rate of all-cause death (17.9% device versus 17.1% control).⁶ Three-year outcomes from the single-arm TRILUMINATE study (n = 98) included a reduction of at least 1 TR grade in 92% of subjects with evaluable transthoracic echocardiography studies (n = 48) and moderate or less TR in 21%.¹⁰ The proportion of individuals with NYHA class III or IV HF symptoms decreased from 76% at baseline to 19% at 3 years. The mean Kansas City Cardiomyopathy Questionnaire score improved by 10 ± 3 points between baseline and 3 years ($P = 0.006$), although it decreased between years 2 and 3. The rate of HFH decreased by 75% comparing events in the 1 year before intervention to events 3 years after T-TEER ($P < 0.0001$).

Directionally similar 1-year results have been reported from the bRIGHT (An Observational Real-World Study Evaluating Severe Tricuspid Regurgitation Patients Treated With the Abbott TriClip Device) single-arm postmarket registry of 511 patients.¹³ The TRI.Fr (Transcatheter Edge-to-Edge Repair for Severe Isolated Tricuspid Regurgitation) trial randomized 300 patients with severe symptomatic TR to T-TEER using the TriClip system plus OMT (n = 152) or OMT alone (n = 148) at 24 centers in France and Belgium.⁹ The primary outcome, a 3-level ordinal clinical composite score at 12 months, was

improved in 74.1% in the T-TEER arm compared with 40.6% in the OMT arm ($P < 0.001$). This benefit was driven primarily by improvements in NYHA class and patient global assessment. At 1 year, the Kansas City Cardiomyopathy Questionnaire score increased by a mean of 15.9 points in the T-TEER group versus 0.4 points in the control group ($P < 0.001$). The proportion of patients with massive or greater TR at 12 months was 6.8% in the T-TEER arm compared with 53.5% in the OMT arm ($P < 0.001$).⁹

The PASCAL system (Edwards Lifesciences) is also designed for T-TEER and has been approved for use in Europe, although not in the United States as of this writing. The PASTE (PASCAL for Tricuspid Regurgitation—a European registry) trial reported outcomes for 1059 patients treated with this system in 16 European centers between 2019 and 2023. Procedural success was achieved in 85%; at 1 year, 83% of evaluable patients showed moderate or less TR with corresponding improvements in functional status and measures of clinical success.³⁰

A significant proportion of patients with severe TR are not candidates for T-TEER because of unfavorable TV anatomy, leaflet coaptation distance, or jet location, among other findings.^{31–33} Some of these challenges can be overcome with TTVR, although this strategy carries its own anatomical, technical, and procedural challenges. EVOQUE (Edwards Lifesciences) is currently the only FDA-approved TTVR device for use in the United States. In the TRISCEND (Edwards EVOQUE Tricuspid Valve Replacement: Investigation of Safety and Clinical Efficacy After Replacement of Tricuspid Valve With Transcatheter Device) single-arm study of 176 patients, the cardiovascular death rate was 1.7% and 9.4% at 30 days and 1 year after EVOQUE implantation, respectively.¹¹ New permanent pacemaker implantation was performed in 13% of patients within 30 days. At 1 year, patients had significant improvement in QoL scores and functional NYHA status. The TRISCEND II Pivotal Trial (Edwards EVOQUE Transcatheter Tricuspid Valve Replacement: Pivotal Clinical Investigation of Safety and Clinical Efficacy Using a Novel Device) was an international, multicenter study in which 400 patients with symptomatic severe TR were randomized (2:1) to either TTVR with the EVOQUE device plus OMT or OMT alone.⁵ Mean age was 79 years, 74% were women, 32% had undergone prior valve surgery or intervention, and 95% had AF. Fourteen percent of patients had primary or CIED-related TR. Treatment with TTVR plus OMT was superior to OMT alone as evidenced by a win ratio of 2.02 (95% CI: 1.56–2.62; $P < 0.001$), an outcome driven primarily by improvements in QoL scores and favorable changes in NYHA class. One-year rates of all-cause death and HFH did not differ between groups. Echocardiographic core laboratory-assessed TR severity was reduced from severe (or greater) to mild or less in

95.3% of patients treated with TTVR. Rates of 30-day all-cause death (3.5%), severe bleeding (10.4%), and new permanent pacemaker implantation (24.7%) with TTVR underscore the need for careful patient selection and informed decision-making. Other adverse events associated with the use of the EVOQUE system in real-world practice reported to the FDA MAUDE (Manufacturer and User Facility Device Experience) database include valve malposition, embolization, and migration, as well as delivery system malfunction. Incidence rates, however, are unavailable.³⁴

The presence of a CIED lead poses particular challenges to the performance of TTVI (more so for TTVR than T-TEER) and requires collaboration with a device expert to advise on lead management before and after intervention. Issues regarding the indications for and risks versus benefits of lead extraction, the implications of “jailing” the lead during TTVR, and the need for alternative pacing systems (eg, leadless, coronary sinus, epicardial) should be considered.

MDT decision-making for patients with valvular heart disease has relied in part on the use of risk scoring systems, particularly the STS-Predicted Risk of Mortality score. Comparable risk scores specific to TTVI are in development and require more refinement as clinical experience increases. The TRIVALVE score (A Risk Score for Mortality/Hospitalization Prediction in Patients Undergoing Transcatheter Tricuspid Valve Intervention) to predict 1-year death rate or HFH is derived from an international registry of 483 patients who had undergone TTVI between 2016 and 2022 (84% T-TEER; 8% TTVR) with various devices.³⁵ The score uses 5 easily measured variables and demonstrates modest discrimination, sensitivity, and specificity. Incorporation of both surgical and transcatheter TV intervention risk scores into MDT deliberations and patient SDM can be considered. It is acknowledged that available risk scores are predictive of hospital-based and 1-year outcomes; data on longer-term outcomes will help place TV interventions into sharper perspective.

1.2. Process

The characterization of quality using the Donabedian triad of structure, process, and outcomes measures is a reasonable framework for establishing quality of care for patients with TR.³⁷ The 2006 Institute of Medicine's report on “Performance Measurement: Accelerating Improvement” provides the context for translating assessment of performance in health care into measures of quality.³⁸ The measures are related to the key periods of patient evaluation, which include patient engagement and decision-making, procedure performance, post-procedural care, and assessment of intermediate- and longer-term outcomes. This document incorporates

several measures into proposed operator and site requirements for a TTVI program for patients with severe native TR.

The focus herein is to optimize quality outcomes across all TTVI sites while providing guidance regarding methods for program assessment. It is likely to be modified by future developments in the field and accumulated outcome data. The writing committee does not consider the recommendations contained herein to exceed the capabilities of most centers or prohibit additional sites from initiating TV programs. The infrastructure and case volumes required to support an active TV surgical program constitute reasonable metrics by which to assess a site's capability to provide TTVI. Emphasis is placed on MDT functionality, interventional echocardiographic expertise, competence in multimodality imaging, technical skill, and transparent reporting. TTVI procedures are inherently more complex than either aortic or mitral valve interventions and necessitate careful consideration of the programmatic components needed to ensure best outcomes.

2. METHODS

The evidence base supporting the recommendations contained in this document is still evolving. The recommendations are based on available trial and registry data, best practices, and expert consensus among the writing committee members representing the professional societies. This guidance document will need updating as TTVIs continue to mature, technology advances, experience accumulates, and more evidence becomes available.

Partnering societies used the ACC's policy on relationships with industry and other entities to author this document.³⁹ To avoid actual, potential, or perceived conflicts of interest that could arise as a result of industry relationships, all writing committee members, as well as peer reviewers of the document, were asked to disclose all current health care-related relationships, including those existing 12 months before initiation of the writing effort. Additionally, writing committees must adhere to the ACC's relationships with industry and other entities (RWI) policy to maintain the appropriate balance of RWI with least 50% of writing committees, plus the chair, not having relevant RWI. Relevant relationships are vetted by the ACC Solution Set Oversight Committee. Author and peer reviewer RWI pertinent to this document are disclosed in [Appendixes 1 and 2](#), respectively. In addition, to ensure complete transparency, the authors' comprehensive disclosure information (including RWI not pertinent to this document) is available as an online [Supplemental Appendix](#). The work of the writing committee was supported exclusively by the partnering societies

without commercial support. Writing committee members volunteered their time for this effort. Conference calls of the writing committee were confidential and were attended only by committee members and relevant staff.

2.1. Abbreviations

Abbreviation	Meaning/Phrase
AF	atrial fibrillation
CED	coverage with evidence development
CIED	cardiac implantable electronic device
HF	heart failure
HFH	heart failure hospitalization
MDT	multidisciplinary team
M-TEER	mitral transcatheter edge-to-edge repair
OMT	optimal medical therapy
PPM	permanent pacemaker
QoL	quality of life
RCT	randomized controlled trial
RV	right ventricular
SDM	shared decision-making
TEE	transthoracic echocardiography
TLE	transvenous lead extraction
TR	tricuspid regurgitation
T-TEER	transcatheter tricuspid valve edge-to-edge repair
TTVI	transcatheter tricuspid valve intervention
TTVR	transcatheter tricuspid valve replacement
TV	tricuspid valve

3. PATIENT EVALUATION AND SELECTION

Suitability for TV intervention is to be evaluated and documented collaboratively by the MDT. In some cases, these discussions will lead to a recommendation against intervention. The MDT collaboration includes discussions among numerous specialists. For accurate diagnosis, prognosis, and effective planning of TV intervention, it is essential to perform multimodality cardiac imaging that specifically targets the anatomy and function of the TV apparatus, right-sided chambers, and adjacent structures^{16,17,40-43} while assessing pertinent left-sided heart disease and pulmonary artery pressure.

In most cases, a general or HF cardiologist refers a patient with severe TR to the MDT. For severe TR, initial review should verify that guideline-recommended management for left-sided heart disease, AF, and/or CIED performance has been established and optimized. The HF specialist can provide recommendations for advanced therapies when indicated. Similarly, secondary TR attributable to AF or CIED-related TR should warrant

involvement of an electrophysiologist for rhythm control and lead management. If severe TR remains despite medical and device therapy optimization and either frailty or comorbidity burden is not considered prohibitive, then referral for consideration of valve intervention, including candidacy for TV surgery or transcatheter treatment, is made. A dedicated interventional echocardiographer should review all imaging data to determine the mechanism(s) of TR, confirm its severity, evaluate imaging features with potential prognostic importance (eg, RV strain, RV-pulmonary artery coupling), and assess the feasibility of surgical and transcatheter options. Direct patient evaluation by the key members of the MDT can be done jointly in a heart valve clinic, both to maximize convenience for patients and facilitate collaborative review, particularly given the comorbidities and complexities of this population. At a minimum, patients should be seen in consultation by an interventional cardiologist and a cardiac surgeon, with imaging reviewed by an interventional echocardiographer. Consensus treatment recommendations should be provided in an SDM process with patients and families, with review of the risks and benefits, including anticipated outcomes with respect to amelioration of HF symptoms, improvement in QoL indicators, expected longevity, and procedural complications.

The risks associated with isolated TV surgery should be documented and may serve as a benchmark against which TTVI can be compared.^{26,27,44,45} Once the MDT decision for transcatheter therapy is made, the TRIVALVE score can be calculated and considered for TTVI risk assessment.³⁵ Major risks with TTVI include death, bleeding, and the need for permanent pacemaker implantation, especially with TTVR. Age, comorbidities, RV function, AF, left-sided heart disease, and residual TR are major drivers of outcomes after TTVI. The risks associated with TTVI should also be documented.

3.1. Multidisciplinary Team

The MDT is a foundational element of the institutional requirements for the establishment of transcatheter programs for the treatment of patients with aortic stenosis and mitral regurgitation.⁴⁶⁻⁴⁸ The “2020 AHA/ACC Guideline for the Management of Patients With Valvular Heart Disease” include a Class 1 recommendation for evaluation by an MDT when intervention is considered.¹ In addition, referral to a comprehensive heart valve center⁴⁷ is preferred when treatment options are being considered for asymptomatic patients with severe valvular heart disease, patients who may benefit from valve repair versus valve replacement, and for patients with multiple comorbidities.

Patients with severe TR often present with multiple comorbidities that should be considered by the MDT.

TABLE 2 MDT Member Requirements for TTVI

General/valve cardiologist
Heart failure specialist
Interventional echocardiographer
Multimodality imaging specialist (with CT and CMR expertise)
Interventional cardiologist
Cardiac valve surgeon
Electrophysiologist, CIED expert
Cardiac anesthesiologist
Advanced practice providers
Program navigator
Structural heart coordinator
Data manager
Other subspecialty expertise as clinically appropriate*

*May include, but not be limited to, the following: geriatricians, nephrologists, hepatologists, hematologists, palliative care specialists, nutritionists, rehabilitation experts, and social workers.

CIED = cardiovascular implantable electronic device; CMR = cardiovascular magnetic resonance; CT = computed tomography; MDT = multidisciplinary team; TTVI = transcatheter tricuspid valve intervention.

Minimum requirements for TV MDT membership (Table 2) include a general or valve cardiologist, HF specialist, interventional echocardiographer, multimodality imaging specialists (eg, echocardiography, computed tomography, cardiac magnetic resonance), interventional cardiologist, cardiac valve surgeon, electrophysiologist/CIED expert, cardiac anesthesiologist, advanced practice providers, program navigator, structural heart coordinator, and data manager. Additional specialists, including geriatricians, nephrologists, hepatologists, hematologists, palliative care specialists, nutrition counselors, rehabilitation experts, and social workers, are included in specific circumstances.

3.1.1. Knowledge Base and Skills of the MDT

The roles of the general cardiologist and HF expert have been described previously. The interventional echocardiographer is expected to evaluate the mechanism and confirm the severity of TR and to consider anatomic and functional criteria that may determine the appropriateness of transcatheter or surgical therapy. These anatomic and functional criteria, inclusive of information derived from cardiac computed tomography or cardiac magnetic resonance, are discussed in detail by the members of the MDT. Integration of imaging data across modalities is essential for procedural planning. The role of the interventional echocardiographer in determination of procedural eligibility, patient selection, and procedural guidance cannot be overemphasized.^{49,50}

The physicians who are present during the performance of the TV procedure will demonstrate the knowledge and skills necessary to accomplish the technical aspects of transcatheter or surgical device deployment

safely and effectively. This includes the skills required of the primary catheter and wire operators (interventional cardiologist, cardiac surgeon, or both), as well as those required of the interventional echocardiographer with respect to 3-dimensional TEE acquisition and interpretation, in addition to guidance during device deployment. However, substantial understanding of the entire procedure, timing of events, technical challenges, and decision points are required of each operator (catheter/wire and imaging) to ensure optimal procedural and patient outcomes.

Assessment by an electrophysiologist is often required when considering TTVR or T-TEER in patients with heart rhythm disorders, CIEDs, or both. Evaluation should address potentially correctable drivers of TR, including AF or chronic left ventricular dysfunction attributable to a wide QRS or a high burden of premature ventricular contractions. In addition, patients with slow AF or pre-existing left bundle branch block may also have a higher risk of needing a permanent pacemaker post-TTVR. CIED lead-related TR may occur in up to 45% of patients, often from mechanical interactions such as leaflet perforation, impingement, adhesion, or interference with the subvalvular apparatus (CIED-related causative TR).⁵¹ Chronic RV pacing, which can lead to pacing-induced cardiomyopathy, is another recognized cause of TR (CIED-related incidental TR).¹⁶ 3-dimensional TEE is recommended to evaluate potential mechanical interactions between the lead and TV apparatus and can provide precise visualization of lead position in >90% of cases.⁵² If CIED-related TR is identified, referral to a device specialist is recommended to assess the risks and benefits of transvenous lead extraction (TLE) and for postprocedural management. Risks associated with TLE (eg, bleeding, perforation with tamponade, leaflet avulsion, need for sternotomy) are low and largely dependent on operator and center expertise and lead dwell time, with newer or more recently placed systems generally easier and safer to remove. Because TLE risks are very low for CIED leads with dwell times of approximately ≤ 2 years, these leads should be routinely removed in preference to leaving them in place if doing so exposes them to any possibility of “jailing.”^{53,54} Dwell time and the risk of TLE exist along a continuum. TLE may also allow for consideration of alternative pacing options if indicated.

Nevertheless, the potential benefit of TLE to improve CIED-related TR is often unpredictable. T-TEER may be feasible despite the presence of a CIED lead, but a CIED lead will become entrapped or “jailed” by a TTVR device. Such lead entrapment precludes future TLE in the event of infection and may therefore limit definitive management options and potentially lead to increased morbidity and death rates. Entrapment also poses risk to the lead's integrity, with dysfunction occurring in 10% to 31% of

patients, a particular concern for pacemaker-dependent patients and for defibrillator leads.^{55,56} Management of CIED leads at the time of surgical TV intervention can include repositioning, lead extraction, and placement of an epicardial system.

Facilities should ensure rapid access to a cardiac electrophysiologist with expertise in lead management to support patients undergoing TTVR. The incidence of new conduction disturbances requiring a permanent pacemaker (PPM) or CIED after TTVR in patients without a preexisting device was 28% at 1 year in the randomized TRISCEND II Pivotal Trial.⁵ T-TEER is not associated with an excess risk of PPM implantation.⁴ Patients with a high risk of postimplant conduction system disease should be referred to an electrophysiologist to consider preemptive PPM placement or to facilitate close short- and long-term follow-up after the TV procedure. Additionally, institutional protocols for prompt temporary and PPM placement if needed post-TTVR should be established. This may include LV pacing via a coronary sinus lead, leadless PPMs, or transvenous leads that cross the TTVR. In addition, it may be helpful to establish institutional protocols for routine assessment of jailed leads post-TTVR.⁵⁷ The potential need for a PPM should also be considered for patients undergoing TV surgery, especially those undergoing TV annuloplasty at time of mitral valve surgery, which might include an epicardial system at the time of surgery if the expected risk of PPM is elevated.^{58,59}

3.1.2. Structure and Function of the MDT

The MDT should meet regularly (preferably weekly) to review case histories, imaging findings, and other pertinent studies, determine procedural risk, achieve consensus on management recommendations including procedure type, review outcomes, assess quality, and verify reporting. The patient's individual needs and preferences should be emphasized. Ideally, the MDT approach will leverage the combined experience and expertise of the group and lead to more standardized and evidence-based decision-making.

Members of the MDT oversee all aspects of evaluation and management, including patient selection, recommended procedure type, preprocedural optimization, procedural performance, postprocedural care, long-term follow-up, outcome reporting, and quality assessment. Optimal outcomes are dependent on a well-coordinated MDT. The MDT should function to minimize the physical, social, and financial burden to the patient.

Formalized reviews of morbidity and death rates, long-term outcomes, quality measures, the STS/ACC TVT Registry and STS Adult Cardiac Surgery Database, and emerging evidence that may influence best practices should be regularly discussed by the MDT. Because TTVIs

have been approved to improve health status patients with severe TR, processes and resources should be allocated to ensure that data on functional status and QoL are collected and publicly reported after intervention. The latter will be particularly important to meet the 24-month data collection and analytic requirements proposed by CMS per the CED study criteria.

3.2. SDM Requirements

Advances in transcatheter heart valve interventions have occurred rapidly with promised advantages over the standard of care, but procedural novelty often introduces uncertainties regarding comparative risks and benefits and long-term outcomes. The complexity of decision-making for patients and clinicians has necessitated guideline recommendations endorsing the adoption of SDM to promote high-quality, patient-centered decisions. The “2020 ACC/AHA Guideline for the Management of Patients With Valvular Heart Disease” emphasizes that “management, in accordance with guideline recommendations, is effective only when followed by both practitioners and patients. Adherence to recommendations can be enhanced by SDM between clinicians and patients, with patient engagement in selecting interventions on the basis of individual values, preferences, and associated conditions and comorbidities.”¹ To this end, patients should: 1) be well-informed regarding all their treatment options and whether all options are available locally; 2) understand the risks and benefits presented using data on treatment options that are as patient-specific as possible; 3) articulate their treatment- and recovery-related goals; 4) express their preferences and values relative to their care; and 5) integrate these to make a final treatment choice. ACC has recently released an informational patient handout for TR.⁶⁰

CMS has similarly emphasized the importance of SDM to enhance beneficiary engagement and incentives: “CMS identifies strengthening beneficiary engagement as one of the agency’s goals to help transform our health care system into one that delivers better care, smarter spending, healthier people, and puts individuals at the center. Specifically, the CMS Quality Strategy envisions health and care that is person-centered, provides incentives for the right outcomes, is sustainable, emphasizes coordinated care and SDM, and relies on transparency of quality and cost information.”⁶¹

Implementation of SDM and consistent achievements of its goals has been challenging. CMS noted, “practitioners have found it difficult to integrate SDM into their routine workflows for various reasons such as overworked physicians, insufficient practitioner training, inadequate clinical information systems, lack of consistent methods to measure that SDM is taking place, and

uncertainty as to whether, or how, to promote change and invest in the time, tools, and training required to achieve meaningful SDM.”⁶² However, SDM is of utmost importance in guiding the use of novel interventions, especially when procedural risks are to be balanced against improvements in QoL, as with TV intervention. Visual decision aids specific to tricuspid valve interventions should be used to facilitate SDM for decisions for or against intervention as well as for the selection of treatment modality (transcatheter versus surgical; replacement versus repair).

3.3. Facilities and Institutional Requirements

Any institution offering a tricuspid valve intervention program should have comprehensive diagnostic imaging services, therapeutic facilities, and an active cardiac surgical program. This includes participation in the STS Adult Cardiac Surgical Database with overall outcomes meeting or exceeding national standards, and the presence of at least 2 institutionally based cardiac surgeons experienced in treating valvular heart disease. Only institutions with well-established interventional cardiology programs should offer TTVI. These programs should include expertise in procedures such as percutaneous coronary intervention, transcatheter aortic valve implantation, M-TEER, balloon valvuloplasty, catheter closure of periprosthetic leaks, and deployment of septal closure devices. These requirements align with the “2019 AATS/ACC/SCAI/STS Expert Consensus Systems of Care Document: Operator and Institutional Recommendations and Requirements for Transcatheter Mitral Valve Intervention.”⁴⁸

These facilities and equipment are essential for a tricuspid valve intervention program:

- Accredited cardiac catheterization laboratory or hybrid operating room/catheterization laboratory equipped with:
 - Fixed radiographic imaging system featuring flat-panel fluoroscopy to provide catheterization laboratory-quality imaging. All participants in the procedure (including interventional echocardiographer, cardiac anesthesiologist, and catheter-wire operator), must have adequate physical shielding from ionizing radiation, as well as institutional monitoring of measured exposures.
 - Sterile environment and adequate space to accommodate equipment for uncomplicated procedures including high-definition monitors, resuscitation cart, oxygen supply, suction, compressed air, CO-oximeter, activated clotting time analyzer, anesthesia equipment, and echocardiography machine.
 - Large variety of interventional equipment including access kits, endovascular sheaths, and introducers

(4- to 26-F); a wide range of guidewires; cardiac diagnostic and interventional catheters; vascular closure devices; balloon dilatation catheters (2-30 mm in diameter) and of various lengths and profiles; a full inventory of coronary and peripheral stents, including covered stents, occlusive vascular devices, snares, and other retrieval devices; drainage catheters; portable vascular access ultrasound; and various implantable device sizes with their delivery systems.

- Access to 3-dimensional intracardiac echocardiography is strongly recommended.
- Mobile, temporary transvenous pacing devices that can be deployed rapidly and can serve as a bridge to a PPM.
- Access to personnel and equipment to implant a PPM in the same catheterization laboratory or facilitate rapid transfer to the electrophysiology laboratory with temporary pacing support. PPM options should include leadless pacing devices and systems that use leads confined to the coronary sinus to avoid crossing a freshly repaired or replaced tricuspid valve.
- Ability to accommodate cardiopulmonary bypass, extracorporeal membrane oxygenation equipment, or both during rescue is recommended.
- Noninvasive cardiovascular imaging programs including:
 - Accredited echocardiography laboratory with skilled sonographers, level 3-trained and National Board of Echocardiography-certified echocardiographers, and at least 1 echocardiographer with specialized expertise in structural interventions with capability for acquisition and quantitative interpretation of transthoracic echocardiography, TEE, and 3-dimensional TEE in patients with tricuspid valve disease.
 - Vascular laboratory (noninvasive) staffed with specialists proficient in performing and interpreting vascular studies.
 - Computed tomography laboratory with a multi-detector scanner (at least 64 slice), staffed by skilled technologists and cardiac imaging specialists.
 - A magnetic resonance imaging laboratory with skilled technologists and cardiac imaging specialists is desirable.
- Postprocedural recovery and intensive care facilities with telemetry monitoring, staffed by physicians, advanced practice providers, and nurses trained and experienced in managing patients who have undergone transcatheter valve procedures.
- Outpatient facilities including office space adequate to accommodate medical, nursing, and technical personnel and ancillary testing facilities (eg,

pulmonary function laboratory, clinical laboratory, imaging facilities) that can provide timely, high-quality care.

- Support services for coordination of postdischarge care (eg, case management, HF program, cardiac rehabilitation).

A successful TTVI program requires institutional support, including financial resources and protected time for participating clinicians. Regularly scheduled MDT conferences and ad hoc interdisciplinary discussions are essential for building consensus on management plans for medically complex patients. Clinicians must dedicate significant time to preprocedural evaluations, patient and family discussions to facilitate SDM, and postprocedural care, much of which may not be reimbursable. Institutions should, whenever possible, view administrative and quality-improvement efforts as strategic investments in long-term programmatic success.

3.4. Outcome Requirements

Unlike other transcatheter valvular interventions, surgical treatment comparison for TTVI are unavailable. The role of medical therapy is limited. As noted, recent STS Adult Cardiac Surgery Database data suggest a death rate of 5.6% and combined mortality and death rate of 26% to 28% at 30 days for isolated surgical treatment.²⁶ The principal outcome measure for surgical TV repair or replacement is a composite score that consists of: 1) risk-adjusted operative death rate (death occurring during the index hospitalization or after transfer to another acute care facility within 30 days, or death within 30 days of surgery); and 2) the risk-adjusted occurrence of any of these 5 major complications: renal failure, stroke, cardiac reoperation, sternal infection/mediastinitis, or prolonged ventilation.²⁶ This composite measure relates to the fundamental short-term goals of surgical procedures for TR, namely, survival with minimal perioperative complications.

Outcome data from observational registries and a landmark RCT demonstrate that T-TEER is safe when performed by skilled operators at experienced centers with low risks of procedural morbidity and death. Risks associated with TTVR are higher (see [Section 1.1](#)). It will be critical to continue to track safety endpoints with TTVIs because these breakthrough technologies evolve and their use becomes more widespread. Longitudinal outcomes and major adverse events should continue to be tracked. Rehospitalization rates can also provide insights on broader aspects of care, including patient selection, procedure performance, and postprocedural care. Other outcomes included in the CMS National Coverage Decisions have been listed previously (see [Preamble](#)).

TABLE 3 Transcatheter TTVI Program Outcome and Performance Monitoring Metrics: Proposed Minimum Quality Standards

Preprocedural process metrics	MDT assessment and consensus recommendation with documentation of: 1. Etiology and severity of TR, HF symptoms, and NYHA class 2. Medical therapies for HF and optimization over ≥ 3 mo by a general cardiologist or (preferably) an advanced HF subspecialist 3. EP specialist assessment of indications for and appropriate initiation of CRT for LV dysfunction with wide QRS (if present) or AF management 4. Surgical risk assessment using STS calculated risk for isolated TV surgery stratified by low ($<4\%$), intermediate (4%–8%) or high ($>8\%$) risk of death.
In-hospital and 30-d all-cause death*	"As expected," "better than expected," or "lower than expected" based on national benchmark data with 95% CIs
Procedural conversion	Conversion from transcatheter repair or replacement to surgical repair or replacement
30-d major vascular complications	95% (outlier) and/or 90% (warning) boundaries on funnel plot
30-d major bleeding	95% (outlier) and/or 90% (warning) boundaries on funnel plot
30-d $\geq 3+$ TR	95% (outlier) and/or 90% (warning) boundaries on funnel plot
30-d significant TS	Mean gradient ≥ 8 mm Hg, valve area <1.5 cm ²
30-d rehospitalization for HF	95% (outlier) and/or 90% (warning) boundaries on funnel plot
30-d improvement in KCCQ score compared with baseline	95% (outlier) and/or 90% (warning) boundaries on funnel plot
30-d improvement in KCCQ score ≥ 10 points compared with baseline	95% (outlier) and/or 90% (warning) boundaries on funnel plot
1-y all-cause death*	"As expected," "better than expected," or "lower than expected" based on national benchmark data with 95% CIs
Change in patient-reported health status (KCCQ score) at 1-y vs. baseline	95% (outlier) and/or 90% (warning) boundaries on funnel plot
1-y rehospitalization for HF	95% (outlier) and/or 90% (warning) boundaries on funnel plot
1-y unplanned repeat TV intervention	Transcatheter or surgical
1-y $\geq 3+$ TR	95% (outlier) and/or 90% (warning) boundaries on funnel plot
1-y significant TS	Mean gradient ≥ 8 mm Hg, valve area <1.5 cm ²

*Risk adjustment model.

AF = atrial fibrillation; CI = confidence interval; CRT = cardiac resynchronization therapy; EP = electrophysiology; HF = heart failure; KCCQ = Kansas City Cardiomyopathy Questionnaire; LV = left ventricle; MDT = multidisciplinary team; NYHA = New York Heart Association; QRS = Quality Rating System; STS = Society of Thoracic Surgeons; TR = tricuspid regurgitation; TS = tricuspid stenosis; TTVI = transcatheter tricuspid valve intervention; TV = tricuspid valve.

Data from the pivotal TTVI trials and registries highlight some of the requirements necessary to maintain a safe, efficient, and effective TTVI program.^{4,5,11,13,35} It appears that different safety thresholds apply for T-TEER versus TTVR. Although the highly selected patient populations enrolled in recent trials at experienced centers may raise concerns regarding generalizability, assessment of safety targets may be possible from ongoing data collection through the STS/ACC TVT Registry and STS Adult Cardiac Surgery Database with linkage to longer-term outcomes. These data reporting requirements should inform quality and safety benchmarks for all sites. Additional and more robust risk-adjusted prediction models for TTVI outcomes will need to be developed once adequate procedural numbers are accrued, similar to those developed for transcatheter aortic and mitral valve interventions. Individual site reports for TTVI procedures can be generated quarterly by the STS/ACC TVT Registry. Consistent with existing precedent for other structural valve procedures, each metric is accompanied by a national benchmark with a median value and interquartile range analyzed from the previous rolling 4 quarters of all site data submitted to the registry.

Several metrics shown in **Tables 3 and 4** can be used to detect outlier performance. For example, worse than expected performance for risk-adjusted measures (once available) on 2 consecutive reports or falling outside the control limits on funnel plots for unadjusted measures should prompt immediate attention and internal quality control interventions. The TV program should consider seeking external review of their performance, including patient selection, procedural success, and quality of care, in the context of the challenges inherent in caring for a patient population that is often elderly and burdened by HF, AF, and additional comorbidities.

3.5. National Registry Requirements

FDA approval of novel transcatheter valve repair or replacement systems does not necessarily guarantee that the device will continue to demonstrate long-term efficacy that is comparable to other treatment options. However, CED remains an extremely powerful mechanism offering significant value to payers, clinicians, and patients. CED has been shown to be an essential technique to permit the rational dispersion of diverse innovative cardiovascular technologies and services into the

TABLE 4 TTVI Program Performance Monitoring Metrics: Proposed Additional Metrics for Quality Assessment

Time Frame	Outcome Metric
In-hospital	■ Technical success* ■ Major vascular bleeding, noncardiac ■ Repeat reintervention for TV disease (transcatheter or surgical procedure that targets new-onset or persistent valve or device dysfunction after a prior TV procedure) or device explant ■ Cardiac complications† ■ Unplanned MCS ■ Cardiac arrest ■ Device embolization ■ Adverse effect on preexisting CIED‡ ■ Venous thromboembolic complication (new DVT, PE, RA, or RV thrombus) ■ Single leaflet attachment (for T-TEER) ■ Paravalvular regurgitation (for TTVR)
30-d	■ All-cause rehospitalization ■ Clinical success§
1-y	■ All-cause rehospitalization ■ Improvement from baseline in symptoms (eg, NYHA improvement by ≥ 1 functional class); and/or improvement from baseline in functional status (eg, 6-min walk test improvement by ≥ 50 m); and/or improvement from baseline in QoL (eg, KCCQ score by ≥ 10 points)

*Per TVARC criteria,¹⁷ for technical success (measured at exit from the procedure room), all of these must be present:

- Absence of intraprocedural mortality or stroke.
- Successful access, delivery, and retrieval of the device delivery system.
- Successful deployment and correct positioning of the intended device(s) without requiring implantation of unplanned additional devices.
- Adequate performance of the transcatheter device. Performance of devices whose purpose is a reduction in TR, should include the absence of tricuspid stenosis (TVA ≥ 1.5 cm² or TVAi ≥ 0.9 cm²/m² [≥ 0.75 if BMI >30 kg/m²], DVI <2.2 , mean gradient <5 mm Hg); reduction of total tricuspid regurgitation to optimal ($\leq$ mild [1+]) or acceptable ($\leq$ moderate [2+]).
- Absence of device-related obstruction of forward flow.
- Absence of device-related pulmonary embolism.
- Freedom from emergency surgery or reintervention during the first 24 h related to the device or access procedure.

†Per TVARC criteria, cardiac complication is defined as

- Cardiac structure perforation, injury, or compromise including but not limited to:
 - Tricuspid valve/leaflet (eg, perforation, tear, restriction, or impingement).
 - Tricuspid annulus/periannular structures (eg, disruption, perforation, or distortion resulting in hemodynamic compromise, atrioventricular conduction disturbance).
 - Right atrium or vena cavae (eg, perforation, aorto-atrial fistula, atrial septal injury).
 - Right ventricle (eg, perforation, pseudoaneurysm, papillary muscle disruption, pulmonic valve injury).
- New pericardial effusion.
- Coronary arterial partial or total occlusion, ischemia.
- Coronary perforation in setting of annular reduction.

‡Per TVARC criteria, adverse effect on preexisting CIED defined as:

- Change in pacing capture threshold (≥ 1 V at 0.5 ms).
- Change in pacing lead impedance (increase or decrease of >200 Ω).
- Change in atrial or ventricular sensing not amenable to reprogramming.
- Lead dislodgement.
- Requirement for device revision after the tricuspid intervention.
- Entrapment of transannular lead by the device.

§Per TVARC criteria, device success at 30 d defined as:

- Absence of procedural mortality or stroke.
- Proper position of the device with adequate performance of the transcatheter device. Performance of devices whose purpose is a reduction in TR should include the absence of tricuspid stenosis (TVA ≥ 1.5 cm² or TVAi ≥ 0.9 cm²/m² [≥ 0.75 if BMI >30 kg/m²], DVI <2.2 , mean gradient <5 mm Hg); reduction of total TR to optimal ($\leq$ mild [1+]) or acceptable ($\leq$ moderate [2+]).
- Freedom from unplanned surgical or interventional procedures related to the device or access procedure.
- Absence of major device or procedure related serious adverse events, including:
 - Life-threatening bleeding (TVARC 5).
 - Major vascular complications.
 - Major cardiac structural complications.
 - Stage 2 or 3 acute kidney injury (includes new dialysis).
 - Myocardial infarction or coronary ischemia requiring percutaneous coronary intervention or coronary artery bypass graft.
 - Device-related obstruction of forward flow.
 - Device-related pulmonary embolism.
 - Any severe hemodynamic compromise leading to heart transplantation or major cardiac assistance or patient extracorporeal membrane oxygenation dependent on day 30.
 - Any valve-related dysfunction, migration, thrombosis, or other complication requiring surgery or repeat intervention.

BMI = body mass index; CIED = cardiac implantable electronic device; DVI = Doppler Velocity Index; DVT = deep vein thrombosis; KCCQ = Kansas City Cardiomyopathy Questionnaire; MCS = mechanical circulatory support; NYHA = New York Heart Association; PE = pulmonary embolism; QoL = quality of life; RA = right atrium; RV = right ventricle; TR = tricuspid regurgitation; T-TEER = transcatheter tricuspid valve edge-to-edge repair; TTVI = transcatheter tricuspid valve intervention; TTVR = transcatheter tricuspid valve replacement; TV = tricuspid valve; TVA = tricuspid valve atresia; TVAi = tricuspid valve area indexed; TVARC = Tricuspid Valve Academic Research Consortium.

marketplace while simultaneously promoting timely clinical safety and effectiveness evaluations. Participation in device-specific registries is essential to the CED process and requires institutional infrastructure and MDT resources that include experienced data managers with a background in cardiovascular disease, funding, office space, computer services, and a data coordinating/clinical research unit with rigorous attention to data precision and accuracy. The importance of obtaining complete 24-month outcomes data, including patient-reported outcomes, deserves further emphasis in view of the CMS National Coverage Decisions.^{7,8} Multisocietal support exists for CED to provide Medicare beneficiaries with prompt access to new technologies and services when early trial evidence suggests benefit, while validating net benefit and risk for beneficiaries in wider real-world cohorts. Clinical registry data are foundational to permit objective assessment of the application of new devices to a more generalizable patient population. Centers that incorporate TTVIs into their practice should participate in the STS/ACC TVT Registry. Mandatory reporting to the STS/ACC TVT Registry under CED through a supplementary module will allow for postmarket surveillance, long-term outcome measurement, and comparative effectiveness research. Merging clinical registry data sets incorporating both TTVI and surgical outcomes would be of interest and may facilitate important research. A TTVI module to supplement the existing STS/ACC TVT modules for transcatheter aortic valve implantation and M-TEER systems will be developed in conjunction with input from specialty societies and relevant stakeholders.

Professional societies have a responsibility to promote long-term data collection and provide collaborative guidance regarding the duration of postmarket surveillance and analysis. Individual centers are responsible for critically evaluating their own program through quality improvement initiatives and for participating in registries to benchmark their performance against national standards. A critical aspect of the integrity of postprocedural surveillance is the need for complete, accurate, and timely data submission from registry sites.

The STS/ACC TVT Registry relies on site-reported data without central echocardiographic core laboratory assessment or regular site monitoring. Nevertheless, several mechanisms are in place to monitor data completeness and accuracy in the STS/ACC TVT Registry. These include site training and support by STS/ACC TVT Registry staff, data “cleaning” by data integrity checks using range validation and other measures, auditing portions of site-level data, and adjudication of selected 30-day and 1-year outcomes. Collection of source documents and verification of prespecified key events can be added specifically for postapproval studies. Both the STS/ACC TVT Registry and STS Adult Cardiac Surgery

Database use a third-party external audit mechanism to assess accuracy and completeness of data submission from sites. Site audits for the STS/ACC TVT Registry began in 2017 using national standards. National registries provide feedback to sites on the quality and completeness of their data submission. Importantly, future comparative analyses may incorporate site-submit data on 1-year outcomes, along with longitudinal survival through linkage with Medicare data, the National Death Index, or both as appropriate.

3.6. Requirements for Establishing and Maintaining a TTVI Program

The establishment of a successful TTVI program often depends on the size and spectrum of the clinical referral base; adequate case volume is needed to ensure viability. The program case volumes contained herein were obtained by querying the STS Adult Cardiac Surgery Database, the STS/ACC TVT Registry, reviewing the published experiences of other data sources, and soliciting expert opinion. The proposed requirements are constructed to: 1) ensure patient safety and promote quality; 2) demonstrate that commitment exists on the part of the institution to the structural heart disease program; and 3) use existing case volume as surrogate for an established TV intervention program to ensure adequate experience, sustainability, and quality. The goal of these recommendations is to promote access to high-quality care. It is self-evident that certain sites would meet these criteria at the outset. Establishment of new sites would follow to expand access and allow for the rational dispersion of new TTVI technologies.

Defining minimum operator and institutional requirements for TTVIs is an important first step to ensure their optimal implementation. The recommended transcatheter/surgical TV competencies are specific to TV interventions but parallel those included in the “2018 AATS/ACC/SCAI/STS Expert Consensus Systems of Care Document: Operator and Institutional Recommendations and Requirements for Transcatheter Aortic Valve Replacement,”⁴⁶ and the “2019 AATS/ACC/SCAI/STS Expert Consensus Systems of Care Document: Operator and Institutional Recommendations and Requirements for Transcatheter Mitral Valve Intervention.”⁴⁸

The writing committee recognizes that CMS did not endorse operator or institutional case volume criteria in their respective memoranda for TTVR or T-TEER coverage under the CED pathway, although it is anticipated that additional studies to clarify “clinical staffing and experience profiles” that associate with patient outcomes in real-world settings will be important.^{6,7} The proposed case volumes for TV intervention pertain to the use of a TTVR or T-TEER device for the improvement of health status in patients with symptomatic, severe TR

TABLE 5 Institutional Criteria for Establishing a T-TEER or TTVR Program

1. ≥ 50 open heart surgeries in the previous year.
2. ≥ 20 tricuspid valve surgeries (isolated or combined) in the 2 y prior.
3. ≥ 2 physicians with cardiac surgery privileges.
4. ≥ 1 physician with interventional cardiology privileges.
5. ≥ 1 physician with interventional echocardiography privileges.
6. 1 board-certified/board-eligible cardiac EP available for pacemaker implantation or lead extraction when required.
7. ≥ 50 TAVI and ≥ 20 TEER procedures per y or ≥ 100 TAVI and ≥ 40 TEER in the previous 2 y.
8. ≥ 200 complete TEE performed per y or ≥ 400 complete TEE performed over the previous 2 y.

EP= electrophysiologist; TAVI = transcatheter aortic valve implantation; TEE = transesophageal echocardiography; TEER = transcatheter edge-to-edge repair; T-TEER = transcatheter tricuspid valve edge-to-edge repair; TTVR = transcatheter tricuspid valve replacement.

despite optimal medical therapy, for whom TV repair or replacement is deemed appropriate by an MDT. The field is rapidly evolving, and recommended case volumes for transcatheter TV repair and replacement systems that may become available in the future may differ. Introduction of this technology across communities should also consider the well-recognized barriers faced by women, under-represented racial and ethnic groups, and persons living in rural or areas remote to surgical and transcatheter cardiac interventions. The case volumes reflect multisociety expert consensus. They are readily achievable for many sites and should not be considered exclusionary.

Ample evidence exists to support an association between higher procedural volumes and better outcomes for transcatheter aortic valve implantation.⁶³⁻⁶⁸ The magnitude and quality of prior institutional and operator M-TEER experience should translate into greater efficiency and intraprocedural communication among the responsible TTVI operators. A recent report from the PASTE T-TEER Registry showed that greater center experience (≥ 21 patients/year) was associated with higher rates of intraprocedural and clinical success.³⁰ The volume-outcome association was also evaluated in a study comparing T-TEER with isolated TV surgery using the Medicare fee-for-service Inpatient Claims and Master Beneficiary Summary Files.⁶⁹ Using a spline analysis, this study showed that 2-year survival after T-TEER appears

to improve up to a total volume of approximately 20 cases but may improve further beyond 60 cases.

To initiate a T-TEER or TTVR program in hospitals without prior TTVI experience, the institutional/hospital program should meet the criteria listed in **Table 5**.

Operator training is a crucial component for treating structural/valvular heart disease using a transcatheter approach (**Table 6**). Construction of a training curriculum encompassing clinical, imaging, procedural and post-procedural elements is essential. Criteria for fellowship and postgraduate training will be established by the participating societies in an expert consensus statement currently under development. After the establishment of a program, all members of the MDT will need to maintain minimal volume requirements for continued participation.

3.6.1. Requirements for the Intraoperative Technical Aspects of T-TEER and TTVR

Requirements for the Intraoperative Technical Aspects of T-TEER

T-TEER should be performed by at least 2 physicians working as cooperators. An interventional echocardiographer must serve as one of these cooperators and provide imaging expertise during all aspects of the procedure. An interventional cardiologist or cardiac surgeon, as the other cooperator, provides expertise in wire, catheter, and device manipulation and deployment. The interventional echocardiographer may not also provide anesthesiology care during the same procedure. At the MDT's discretion, if appropriate, an additional physician, either an interventional cardiologist or a cardiac surgeon, can be involved as a third operator for the procedure. All physicians who participate in the procedure must have device-specific training as required by the manufacturer.

Requirements for the Intraoperative Technical Aspects of TTVR

At the initiation of a TTVR program, 3 physician operators will actively participate in the TTVR procedure:

1. An interventional echocardiographer should be considered a cooperator for TTVR and must provide

TABLE 6 Requirements for Catheter and Interventional Echocardiographer Operators to Establish a T-TEER or TTVR Program

Catheter Operator Requirements	Interventional Echocardiographer Operator Requirements
1. ≥ 50 career structural valve procedures of which ≥ 20 are TEER procedures. ⁷⁰	1. ≥ 50 career structural valve disease procedures, of which ≥ 20 are TEER procedures.
2. Board eligibility, equivalency, or certification in either interventional cardiology or cardiothoracic surgery.	2. Level 3 or equivalent board eligibility or certification in echocardiography.
3. Certification of device-specific training.	3. Certification of device-specific training.

TEER = transcatheter edge-to-edge repair; T-TEER = transcatheter tricuspid valve edge-to-edge repair; TTVR = transcatheter tricuspid valve replacement.

imaging expertise in patient and device selection as well as catheter position and device deployment. The interventional echocardiographer will not also provide anesthesia during the procedure.

2. An interventional cardiologist provides expertise in hemodynamic assessment, vascular access, catheter management, wire manipulation, and device deployment.
3. A cardiac surgeon skilled in transcatheter valve procedures provides additional expertise in catheter management, wire manipulation, and device deployment, as well as intraprocedural complication management.

All physicians who participate in the procedure should be board certified or equivalent in their specialty and have received device-specific training as required by the device manufacturer. All TTVR institutions are to submit complete data to the STS/ACC TVT Registry.

The joint 3 specialty physician composition of the intraprocedural team can be reconsidered by the institution after consistent demonstration of procedural safety with a 30-day major complication rate of <20% (composite of cardiovascular death, severe bleeding, nonelective TV reintervention, and major access/vascular/cardiac structural complications) as documented in the STS/ACC TVT Registry after a minimum of 25 cases or 3 years of TTVR experience. On meeting these safety and quality requirements, the composition of the team may be adjusted to include an interventional echocardiographer (who must provide imaging expertise during all aspects of the procedure and may not also provide anesthesia services), and one other procedural cooperator skilled in transcatheter procedures with device-specific training, who may be an interventional cardiologist or cardiac surgeon with appropriate certification. After achieving this initial minimal safety and quality threshold, TTVR institutions should demonstrate maintenance of this threshold of <20% complication rate as documented by the STS/ACC TVT Registry on an annual basis. Institutions not achieving this minimum annual threshold of safety and quality should revert to the original 3-operator composition until such time that this quality threshold can be maintained for 3 consecutive years.

3.6.2. Qualifications to Maintain a T-TEER or TTVR Program

To optimize outcomes, sites with TV interventional experience should maintain these:

Site Qualifications to Maintain a T-TEER Program

- a) ≥ 20 tricuspid valve interventions (transcatheter or surgical), of which ≥ 10 are transcatheter over 1 year, or ≥ 40 tricuspid valve interventions (transcatheter or

surgical), of which ≥ 20 are transcatheter over 2 years, and;

- b) The initial complement of specialty expertise should be maintained, and;
- c) ≥ 50 transcatheter structural cases per year or ≥ 100 over the previous 2 years.

Site Qualifications to Maintain a TTVR Program

- a) ≥ 20 tricuspid valve interventions (transcatheter or surgical), of which ≥ 10 are transcatheter and of which ≥ 5 are TTVR in 1 year, or ≥ 40 tricuspid valve intervention (transcatheter or surgical), of which ≥ 20 are transcatheter and of which ≥ 10 are TTVR over 2 years, and;
- b) The initial complement of specialty expertise should be maintained, and;
- c) ≥ 50 transcatheter structural cases per year or ≥ 100 over the previous 2 years.

3.7. Compliance With Document Recommendations

The standards outlined in this document are intended to ensure that patients undergoing tricuspid valve interventions receive optimal care. Although compliance with these case volume recommendations will be voluntary, the writing committee believes that these minimums provide reasonable thresholds to ensure quality and enhance the likelihood of meeting the CMS proposed study criteria for CED.

The use of the STS/ACC TVT Registry for tracking transcatheter aortic and mitral valve interventional outcomes is well-established. Similarly, programs' participation in the Registry for TTVI should allow for postmarketing surveillance, long-term outcome measurement, and comparative effectiveness research. Annual Registry reports will summarize the number of sites with active TTVI programs and their procedural volumes, as well as key outcomes. Programs that do not meet standards for volume, quality of care, or reporting of performance or process metrics will be able to launch quality improvement initiatives to address any deficiencies. Collaborative professional society feedback will be instrumental in assisting programs achieve and maintain high standards of care.

CONCLUSIONS

The multisociety writing committee has undertaken this effort to provide guidance for the initiation and maintenance of a TTVI program in a rapidly evolving field. Emphasis is placed on the roles and responsibilities of the MDT and its members to promote optimal patient and procedural selection to ensure safest and highest quality outcomes. Patients who present with symptomatic severe native valve TR generally have advanced HF, AF, and

numerous comorbidities. As noted, TTVIs can be complex and require careful planning. Rigorous data collection and comparative analysis, inclusive of functional status and QoL indicators, will be essential to validate the appropriateness of the continued growth in procedural volumes. Introduction of alternative and newer transcatheter systems is anticipated. It is expected that the criteria proposed herein will evolve with further advances in patient selection, equipment, and both interventional and surgical techniques. Nevertheless, the guiding principles and foundational elements included in this document constitute an enduring commitment to optimizing patient outcomes while acknowledging the need for equitable access.

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KEY WORDS ACC Expert Consensus System of Care, coverage with evidence development, transcatheter edge-to-edge repair, tricuspid regurgitation, transcatheter tricuspid valve intervention, transcatheter tricuspid valve replacement

APPENDIX 1. AUTHOR RELATIONSHIPS WITH INDUSTRY AND OTHER ENTITIES (RELEVANT)—OPERATOR AND INSTITUTIONAL RECOMMENDATIONS AND REQUIREMENTS FOR TRICUSPID INTERVENTIONS: 2026 ACC/AHA/ASE/HRS/STS EXPERT CONSENSUS SYSTEMS OF CARE DOCUMENT

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ACC = American College of Cardiology; AHA = American Heart Association; ASE = American Society of Echocardiography; DICTATE-AHF = Efficacy and Safety of Dapagliflozin in Acute Heart Failure; HRS = Heart Rhythm Society; STS = Society of Thoracic Surgeons.

APPENDIX 2. PEER REVIEWER RELATIONSHIPS WITH INDUSTRY AND OTHER ENTITIES (COMPREHENSIVE)—OPERATOR AND INSTITUTIONAL RECOMMENDATIONS AND REQUIREMENTS FOR TRICUSPID INTERVENTIONS: 2026 ACC/AHA/ASE/HRS/STS EXPERT CONSENSUS SYSTEMS OF CARE DOCUMENT

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*No financial benefit.

†Significant relationship.

ACC = American College of Cardiology; AHA = American Heart Association; APOLLO = A Phase 3 Study Comparing Pomalidomide and Dexamethasone With or Without Daratumumab in Subjects With Relapsed or Refractory Multiple Myeloma Who Have Received at Least One Prior Line of Therapy With Both Lenalidomide and a Proteasome Inhibitor; ASE = American Society of Echocardiography; HRS = Heart Rhythm Society; NIH = National Institutes of Health; PARTNER3 = A Prospective, Randomized, Controlled, Multi-Center Study to Establish the Safety and Effectiveness of the SAPIEN 3 Transcatheter Heart Valve in Low Risk Patients Who Have Severe, Calcific, Aortic Stenosis Requiring Aortic Valve Replacement; STS = Society of Thoracic Surgeons; TRILUMINATE = Trial to Evaluate Cardiovascular Outcomes in Patients Treated With the Tricuspid Valve Repair System Pivotal; TRISCEND II = Edwards EVOQUE Transcatheter Tricuspid Valve Replacement: Pivotal Clinical Investigation of Safety and Clinical Efficacy Using a Novel Device.